

# nccl-tests (sendrecv) 1-node summary

- Generated: 2026-07-13 16:21:48
- Collective: sendrecv\_perf
- Nodes: node5500, node5502 (each 8 x NVIDIA B200, single node, intra-node NVLink)
- Config: 1 thread, 1 MiB-16 GiB, 5 warmup + 20 iters
- Reference (MIT aicr-benchmarks, results\_b200.md, b0027): sendrecv busbw **666 GB/s** = 74% of 900 GB/s NVLink max

## Per-node overview

| Node                     | Avg busbw (GB/s) | Peak busbw (GB/s) | Converged (GB/s) | % of NVLink max | % of reference | Correctness |
|--------------------------|------------------|-------------------|------------------|-----------------|----------------|-------------|
| node5500                 | 318.3            | 665.5             | 665.5            | 74%             | 99.9%          | PASS        |
| node5502                 | 317.2            | 664.9             | 664.9            | 74%             | 99.8%          | PASS        |
| <b>reference (b0027)</b> | —                | 666.0             | 666.0            | 74%             | 100%           | —           |

Converged = busbw at the largest message size (16 GiB), best of out-of-place / in-place (matches the reference's methodology).

## Bus bandwidth vs message size (out-of-place busbw, GB/s)

| Message size | node5500 | node5502 |
|--------------|----------|----------|
| 1 MiB        | 28.6     | 22.6     |
| 4 MiB        | 61.8     | 63.6     |
| 16 MiB       | 78.0     | 77.6     |
| 64 MiB       | 84.9     | 84.4     |
| 256 MiB      | 332.8    | 332.3    |
| 1 GiB        | 645.4    | 643.3    |
| 4 GiB        | 660.8    | 660.4    |
| 16 GiB       | 665.5    | 664.9    |

In-place busbw tracks out-of-place within ~2% for sendrecv. Bandwidth rises with message size and saturates at large sizes (NVLink-bound).